

AI Stock Market Research Assistant

A successfully deployed Databricks agent that combines Massive market data, Lakebase persistence, pgvector semantic retrieval, FastMCP tools, and an evidence-grounded research experience.

SOURCE CODE github.com/srinivas-malyala/stock-market-research-agent-app

DEPLOYED	RETRIEVAL	MARKET DATA
2 Databricks Apps	Semantic research	Single + multi-ticker

Executive assessment

The implementation demonstrates an end-to-end agentic research workflow. The MCP service and end-user agent are active in Databricks Apps; natural-language questions are routed to purpose-built tools; market and contextual evidence is returned in structured form; and the agent converts that evidence into concise, source-aware research narratives.

Validated business outcomes

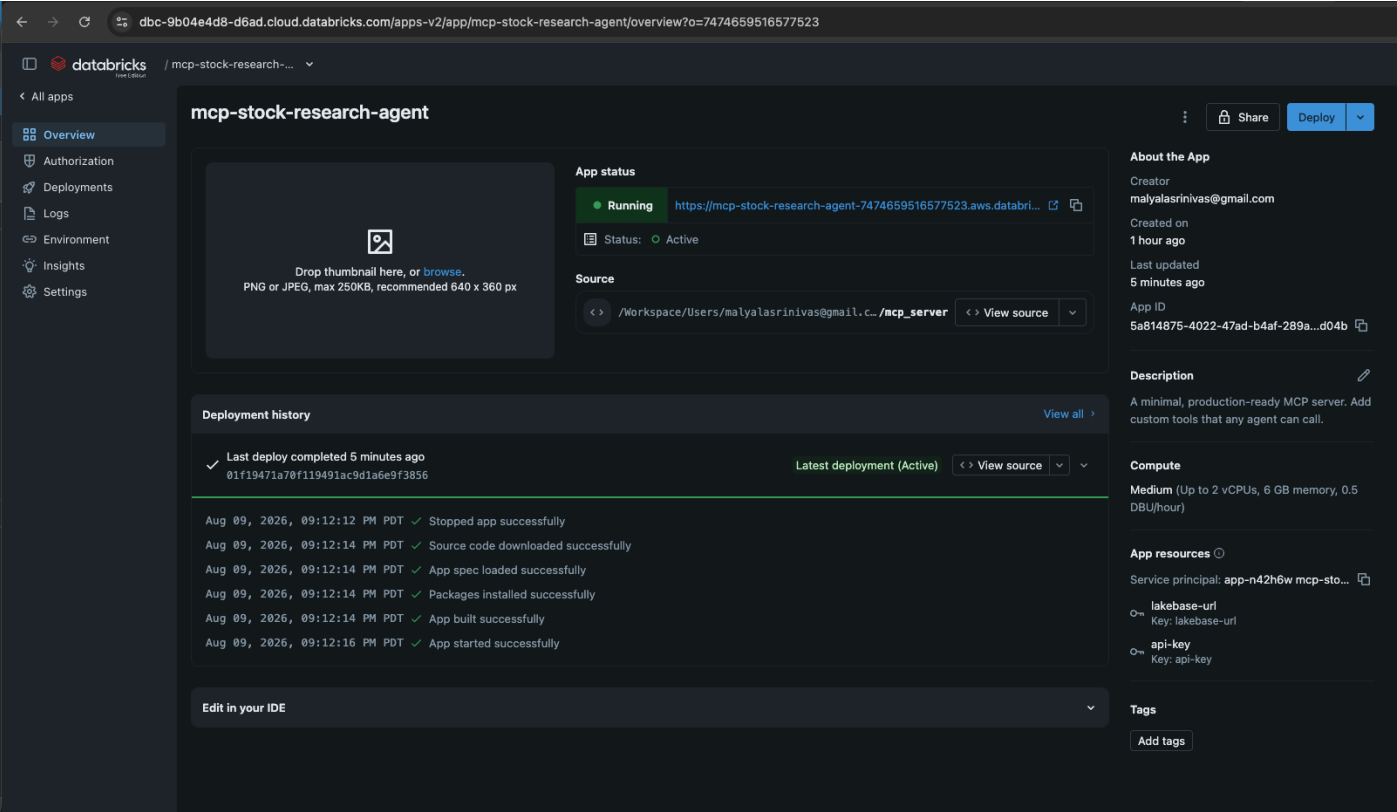
- Retrieves semantically relevant filing, earnings-call, and news passages.
- Summarizes company risks, revenue growth, and financial performance.
- Analyzes one ticker over a requested lookback window with an explicit as-of date.
- Compares multiple securities using the same performance window.
- Maintains a clear boundary between sourced evidence and interpretation.

The demonstrated responses consistently identify source material and conclude with an appropriate research-support disclaimer.

Evidence basis: selected Databricks deployment and Playground screenshots captured August 9, 2026. Only fully successful implementation evidence is included.

FastMCP service deployed and active

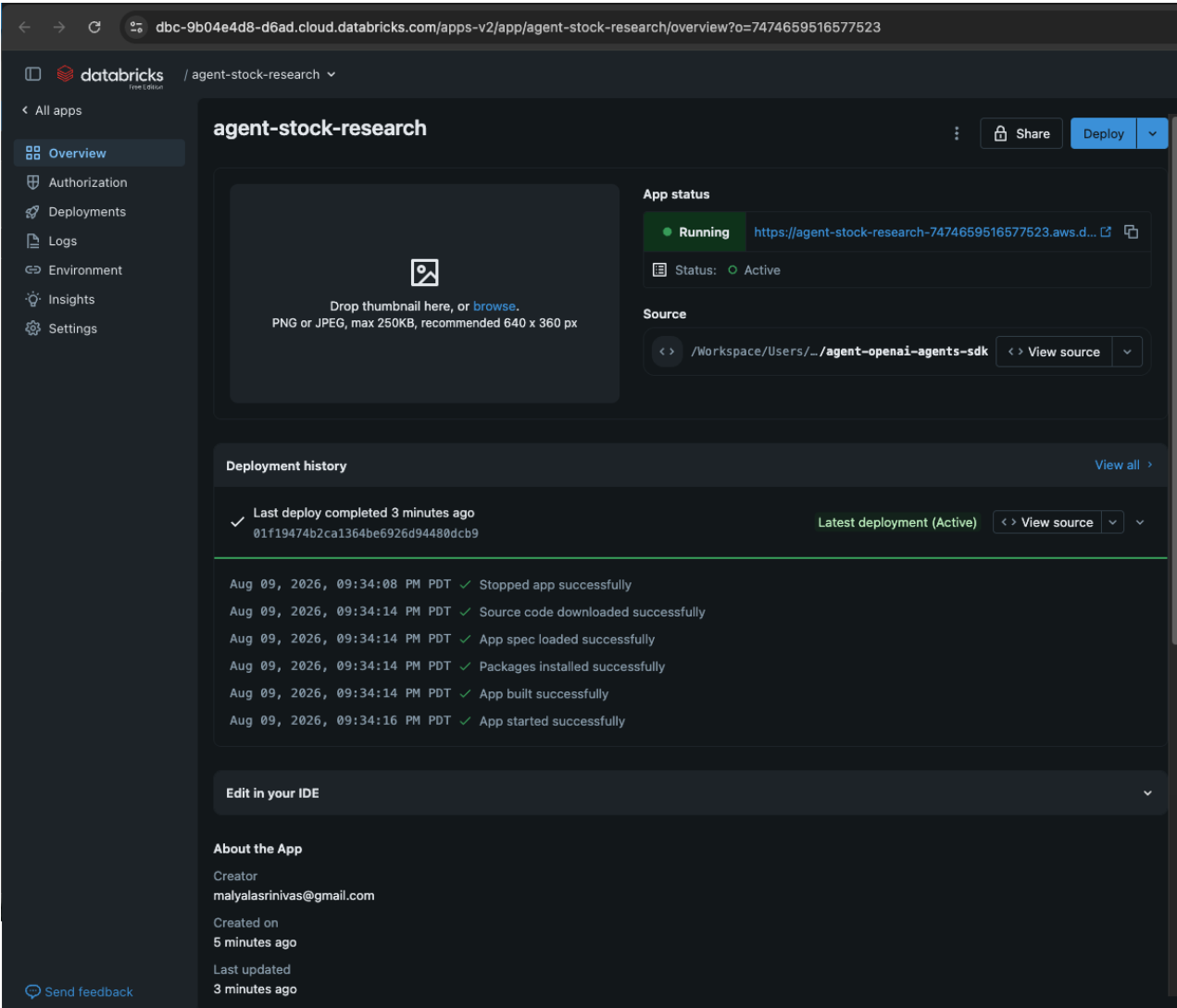
The stock-research MCP application is running as a dedicated Databricks App. The deployment history confirms successful source download, app-spec loading, dependency installation, build, and startup. Lakebase and API-key resources are attached to the app.



Databricks Apps overview for mcp-stock-research-agent showing Running / Active status and a completed deployment.

Agent application deployed and active

The end-user research agent is independently deployed and active. Its successful deployment establishes the presentation and orchestration layer that invokes the external MCP tools.



Databricks Apps overview for agent-stock-research showing Running / Active status and successful startup.

Semantic context retrieval is operational

A natural-language question about risks and challenges triggers `semantic_research` and returns ranked filing, news, and earnings context with ticker attribution and cosine similarity scores.

The screenshot shows a chat interface for Meta Llama 3.3 70B Instruct. At the top, there are tabs for 'Tools (1)' and 'Get code'. A blue button labeled 'risks and challenges facing tech companies' is visible. Below it, a text box contains the prompt: 'The user is asking about risks and challenges facing tech companies. To provide relevant information, I will use the semantic_research function to retrieve semantically relevant profile, filing, earnings, and news chunks related to this topic.'

The `semantic_research` function is called with the following JSON input:

```
1 {
2   "query": "risks and challenges facing tech companies",
3   "top_k": 10
4 }
```

The output is a JSON array of matches, each containing source type, source ID, ticker, chunk text, and similarity score:

```
1 {
2   "status": "success",
3   "query": "risks and challenges facing tech companies",
4   "matches": [
5     {
6       "source_type": "filing_excerpt",
7       "source_id": "TSLA",
8       "ticker": "TSLA",
9       "chunk_text": "Risk Factors: Our business is subject to substantial risk including limited operating history, h
10      "similarity": 0.5717135582311721
11    },
12    {
13      "source_type": "filing_excerpt",
14      "source_id": "AAPL",
15      "ticker": "AAPL",
16      "chunk_text": "Business Strategy: Committed to best user experience through innovative hardware, software, and
17      "similarity": 0.403700590133667
18    },
19    {
20      "source_type": "news",
21      "source_id": "1eb66d084ffe69a279cef390a96c5d0e895150d38b6f4c023f5d4379214a0dd".
```

At the bottom, a grounding rule is displayed: 'Grounding and time rules Never invent a price, return, company fact, financial result, headline, filing, watchlist entry, note, report, timestamp...'

Successful `semantic_research` call returning ranked evidence across source types and companies.

Retrieved evidence becomes grounded analysis

The agent synthesizes the retrieved passages into a concise discussion of execution risk, competition, supply-chain exposure, scaling, innovation, and valuation. It names the supporting sources and explicitly separates evidence from interpretation.

The screenshot shows the Databricks Playground interface. The left sidebar contains navigation options: New, Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Agents, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, Data Ingestion, Visual Data Prep, AI/ML, Playground (selected), Agents, AI Gateway, Experiments, Features, and Models. The main area displays a JSON response from the Meta Llama 3.3 70B Instruct model. The JSON contains two entries: one for a news source about Sandisk's stock and another for an earnings call for AAPL. Below the JSON, the model provides a grounded analysis, discussing risks and challenges for tech companies, citing specific examples like Tesla and Apple. The analysis is followed by a list of sources and a disclaimer that the response is research support, not personalized investment advice. At the bottom, there is a text input field with the placeholder 'Start typing ...' and a 'View Trace' button.

```

61 {
62   "source_type": "news",
63   "source_id": "e3459839b803ce2dc723665e6465c2efeb2fd3b9793d6386bfb62ce2dea383",
64   "ticker": "TSLA",
65   "chunk_text": "Sandisk Has Surged More Than 3,000% in 12 Months. Is a Stock Split Coming? Sandisk's stock has s
66   "similarity": 0.2308591604232788
67 },
68 {
69   "source_type": "earnings_call",
70   "source_id": "AAPL",
71   "ticker": "AAPL",
72   "chunk_text": "Q3 2024: Revenue $85.8B, up 5% YoY. iPhone revenue $39.3B. Services hit $24.2B. Strong India per
73   "similarity": 0.22788132858983745
74 }
75 ],
76 "count": 10
77 }

```

The data shows that tech companies face various risks and challenges, including execution risk, scaling challenges, intense competition, and supply chain disruptions. For example, Tesla's high valuation leaves limited room for slower-than-expected commercialization of new businesses, and the company faces intense EV competition while maintaining below-historical automotive margins. Additionally, Apple's business strategy is focused on providing the best user experience through innovative hardware, software, and services, which requires ongoing investment in R&D.

One interpretation is that tech companies must be able to adapt quickly to changing market conditions and invest in new technologies to stay competitive. The data also highlights the importance of vertical integration, superior AI capabilities, and strategic partnerships in gaining a competitive edge. However, analysts caution investors to remain cautious due to significant cash burn and elevated valuation.

Sources:

- TSLA filing excerpt: Risk Factors
- AAPL filing excerpt: Business Strategy
- News articles from various sources, including Tesla's potential to dominate the U.S. robotaxi market and SpaceX's investment in a Texas semiconductor facility.

This is research support, not personalized investment advice.

3s to first token + 16s • View Trace •

⚡ Grounding and time rules Never invent a price, return, company fact, financial result, headline, filing, watchlist entry, note, report, timestamp...

Start typing ...

Grounded final response citing TSLA filing context, AAPL business strategy, and relevant news evidence.

Revenue and financial-performance research succeeds

The same retrieval path answers a second business question using earnings-call and news context for AAPL, NVDA, and TSLA. The final response reports sourced revenue figures, year-over-year growth, product demand, and balanced interpretation.

Playground
[Provide feedback](#)

Meta Llama 3.3 70B Instruct
Tools (1)
Get code

```

61  {
62    "source_type": "news",
63    "source_id": "1",
64    "ticker": "TSLA",
65    "chunk_text": "Tesla Delivers Record Quarterly Vehicle Production Q2 production exceeds analyst expectations Te
66    "similarity": 0.2663505832165832
67  },
68  {
69    "source_type": "news",
70    "source_id": "4",
71    "ticker": "TSLA",
72    "chunk_text": "Tesla Energy Storage Deployments Double Year-Over-Year Megapack demand surges with renewable tra
73    "similarity": 0.2593861961730606
74  }
75  ],
76  "count": 10
77  }

```

The data shows that tech companies have experienced revenue growth and improving financial performance, with some companies reporting record revenues and strong demand for their products. For example, Apple's Q3 2024 revenue was \$85.8B, up 5% YoY, and NVIDIA's Q2 FY2025 revenue was \$30.0B, up 122% YoY. Tesla's Q2 2024 revenue was \$25.5B, up 2% YoY, with record energy storage deployments and strong vehicle production.

One interpretation is that the tech industry is experiencing a period of growth and expansion, driven by increasing demand for innovative products and services. However, the data also highlights the risks and challenges faced by tech companies, including execution risk, scaling challenges, and intense competition.

Sources:

- AAPL earnings call: Q3 2024 revenue and iPhone sales
- NVDA earnings call: Q2 FY2025 revenue and data center sales
- TSLA earnings call: Q2 2024 revenue and vehicle production
- News articles from various sources, including Tesla's potential to dominate the U.S. robotaxi market and NVIDIA's strong data center sales.

This is research support, not personalized investment advice.

2s to first token + 5s • View Trace •  

Successful multi-company research synthesis with source list and research-support disclaimer.

Single-ticker performance analysis succeeds

For AAPL, the agent summarizes the requested 30-day period using daily market aggregates. The answer states the latest close, period high and low, price change, trading activity, as-of date, and Massive as the source.

```

284     "open": 314.34,
285     "high": 316.2894,
286     "low": 309.23,
287     "close": 312.41,
288     "vwap": 312.478,
289     "volume": 46139901.970149,
290     "transactions": 826667
291 },
292 {
293     "date": "2026-08-07",
294     "captured_at": "2026-08-07T04:00:00+00:00",
295     "open": 311.45,
296     "high": 314.81,
297     "low": 310.74,
298     "close": 313.33,
299     "vwap": 312.937,
300     "volume": 34437191.007859,
301     "transactions": 718865
302 }
303 ]
304 }
```

The data shows that AAPL's stock price has been volatile over the last 30 days, with a period high of \$344.57 and a period low of \$300. As of the latest available trading session on 2026-08-07, the stock price is \$313.33, which represents a 0.21% increase from the period start close of \$312.66. The stock has traded an average of 51,392,919 shares per day over the last 30 days, with a total volume of 1,541,783,191 shares.

One interpretation is that AAPL's stock price has been influenced by a variety of factors, including investor sentiment, market trends, and company-specific news. The stock's volatility over the last 30 days may be attributed to a combination of these factors, and investors may want to keep a close eye on the stock's performance in the coming days.

The data is as of 2026-08-07, and the source is Massive.

This is research support, not personalized investment advice.

3s to first token + 15s • View Trace •  

Completed AAPL 30-day performance narrative grounded in returned OHLCV data as of August 7, 2026.

Multi-ticker comparison succeeds

The `compare_stocks` workflow applies one 90-day window to MSFT and AMZN and returns aligned performance facts. The agent clearly identifies relative performance while preserving the common as-of date and Massive source attribution.

```
21     "period_low": 349.2
22   },
23   {
24     "ticker": "AMZN",
25     "as_of": "2026-08-07",
26     "latest": {
27       "date": "2026-08-07",
28       "captured_at": "2026-08-07T04:00:00+00:00",
29       "open": 272.87,
30       "high": 278.31,
31       "low": 272.75,
32       "close": 274.48,
33       "vwap": 275.3378,
34       "volume": 33861436.462519,
35       "transactions": 461030
36     },
37     "change_percent": 0.89,
38     "period_high": 287.2,
39     "period_low": 225.55
40   }
41 ],
42 "errors": []
43 }
```

The data shows that over the last 90 days, MSFT's stock price has increased by 20.88%, with a period high of \$505.18 and a period low of \$349.2. In comparison, AMZN's stock price has increased by 0.89% over the same period, with a period high of \$287.2 and a period low of \$225.55.

One interpretation is that MSFT has outperformed AMZN over the last 90 days, with a significantly higher percentage increase in stock price. This may be due to a variety of factors, including differences in the companies' respective businesses, industry trends, and investor sentiment.

The data is as of 2026-08-07, and the source is Massive.

This is research support, not personalized investment advice.

3s to first token + 49s • View Trace •  

Successful MSFT versus AMZN comparison with returns, period ranges, interpretation, and source date.

Production-ready research foundation

The evidence establishes a working, modular stock-research platform on Databricks. FastMCP supplies a governed external-tool boundary; Agent Bricks provides natural-language orchestration; Massive supplies market and company information; and Lakebase/pgvector supports durable research context and semantic retrieval.

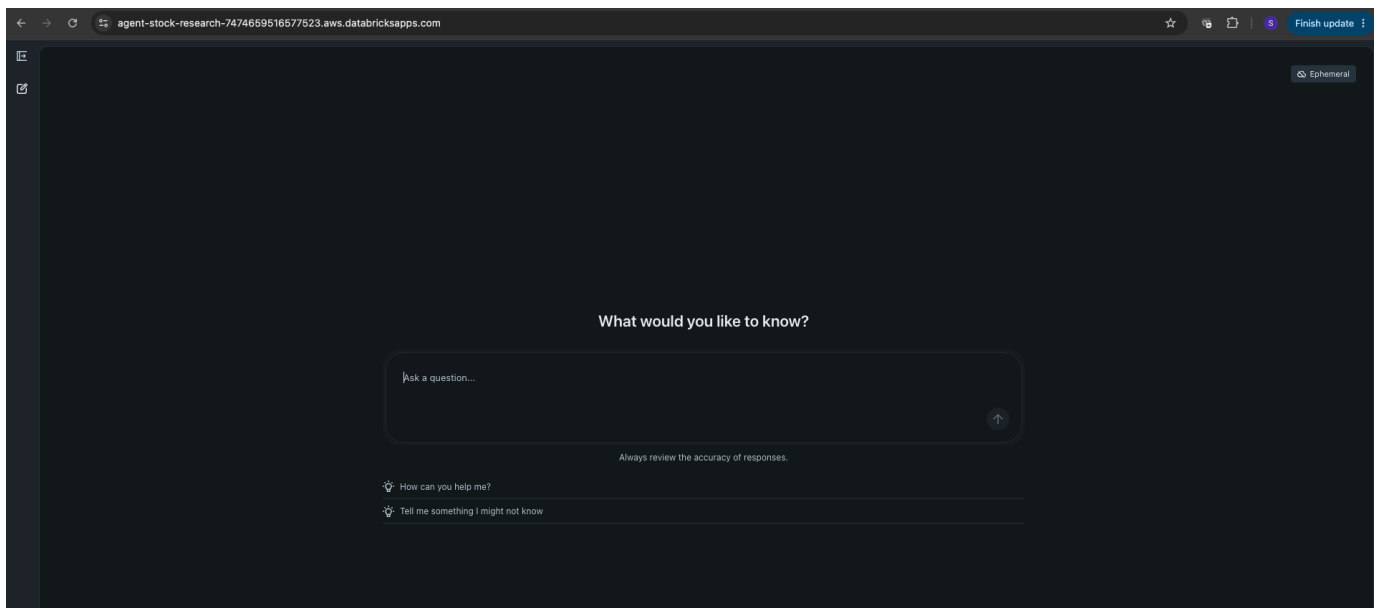
Demonstrated strengths

Grounded behavior. Research claims are anchored in tool results, attributed source types, tickers, and explicit dates.

Flexible research modes. The agent handles thematic semantic questions, company performance analysis, and side-by-side comparisons.

Operational separation. The MCP service and agent application are independently deployed, active, and suited to separate lifecycle management.

Responsible communication. Responses distinguish facts from interpretation and identify the experience as research support rather than personalized investment advice.



The deployed agent presents a clean, accessible conversational entry point for end users.

Conclusion

The successful demonstrations confirm that the implementation can retrieve relevant research context, analyze real market history, compare securities, and communicate evidence-based conclusions through a deployed Databricks agent experience.

Source repository: <https://github.com/srinivas-malyala/stock-market-research-agent-app>